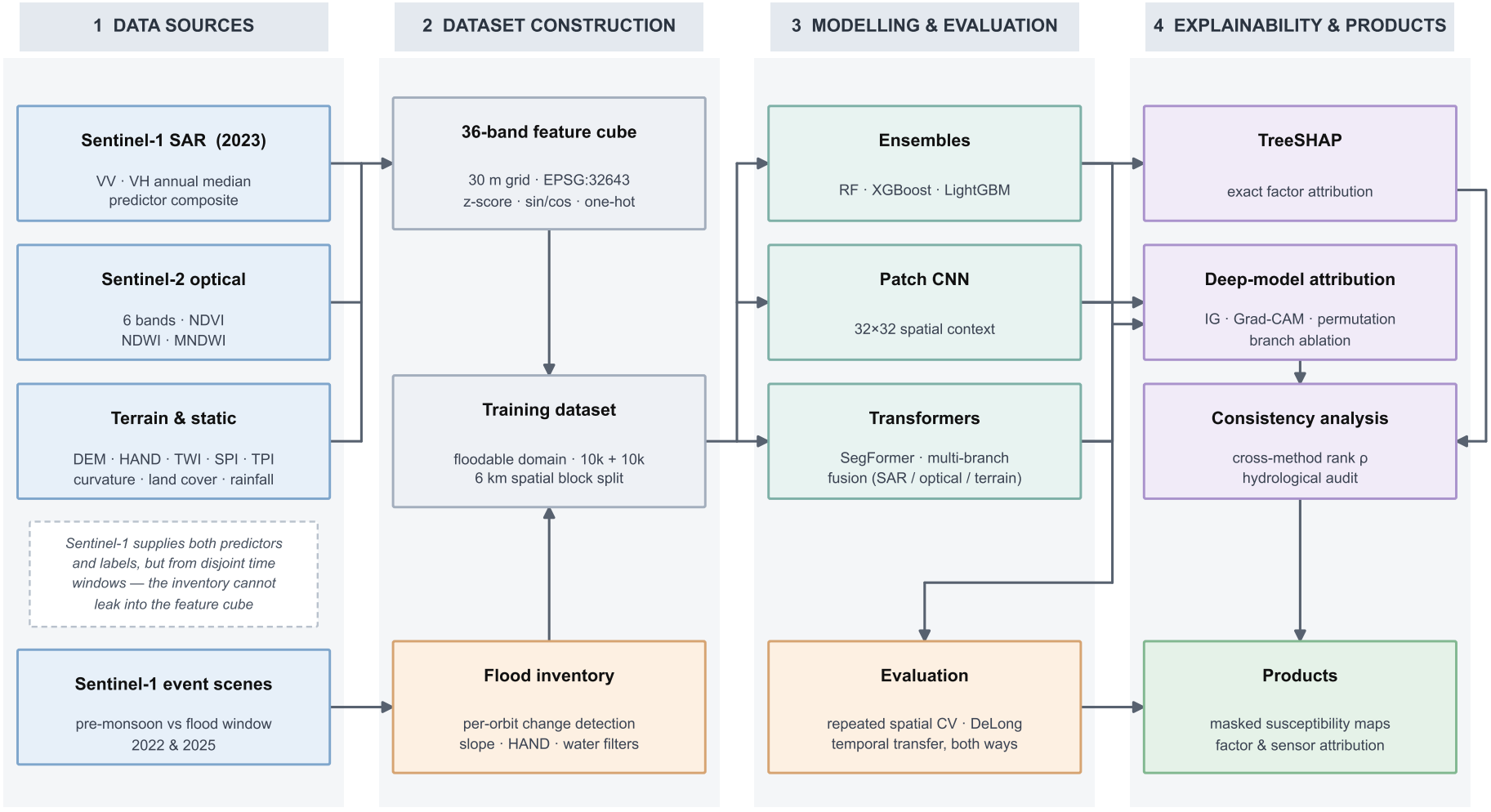

1 DATA SOURCES
2 DATASET CONSTRUCTION
3 MODELLING & EVALUATION
4 EXPLAINABILITY & PRODUCTS
Sentinel-1 SAR (2023)
VV · VH annual median predictor composite
Sentinel-2 optical
6 bands · NDVI NDWI · MNDWI
Terrain & static
DEM · HAND · TWI · SPI · TPI curvature · land cover · rainfall
Sentinel-1 supplies both predictors and labels, but from disjoint time windows — the inventory cannot leak into the feature cube
Sentinel-1 event scenes
pre-monsoon vs flood window 2022 & 2025
36-band feature cube
30 m grid · EPSG:32643 z-score · sin/cos · one-hot
Training dataset
floodable domain · 10k + 10k 6 km spatial block split
Flood inventory
per-orbit change detection slope · HAND · water filters
Ensembles
RF · XGBoost · LightGBM
Patch CNN
32×32 spatial context
Transformers
SegFormer · multi-branch fusion (SAR / optical / terrain)
Evaluation
repeated spatial CV · DeLong temporal transfer, both ways
TreeSHAP
exact factor attribution
Deep-model attribution
IG · Grad-CAM · permutation branch ablation
Consistency analysis
cross-method rank ρ hydrological audit
Products
masked susceptibility maps factor & sensor attribution